Module 4

UVM Environment & Checker Maturity (SV/UVM)

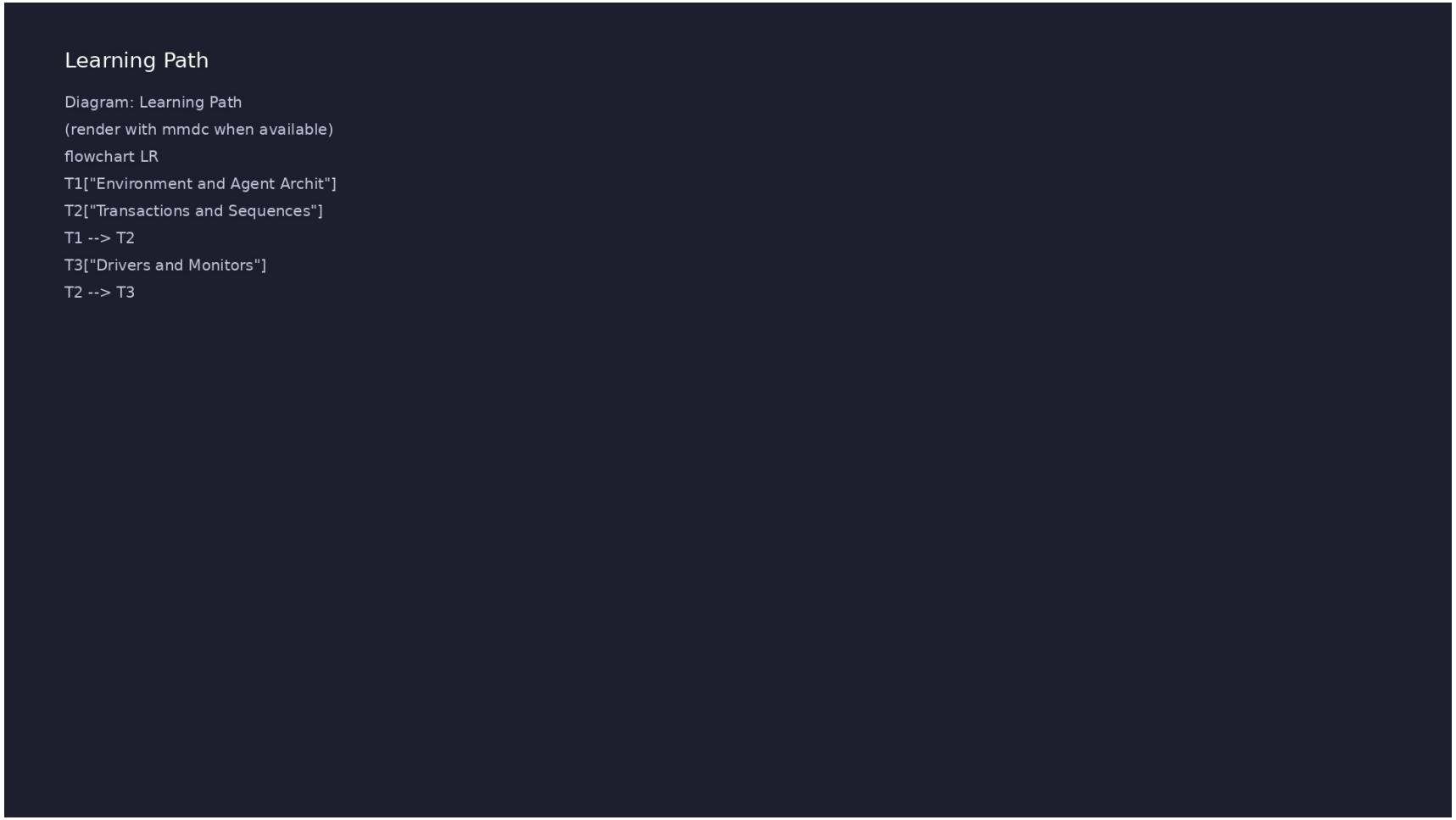
Learning objectives

- Design and mature your **UVM environment** (env, agents, drivers, monitors, sequencers) and **check...

Prerequisites

- See module README

Learning path



Design and mature your **UVM environment** (env, agents, drivers, monitors, sequencers) and **checking strategy** (scoreboards, reference m...

Overview

- Modules 1–3 focused on planning (scope, tests, regressions) and measurement (coverage). In Module 4...

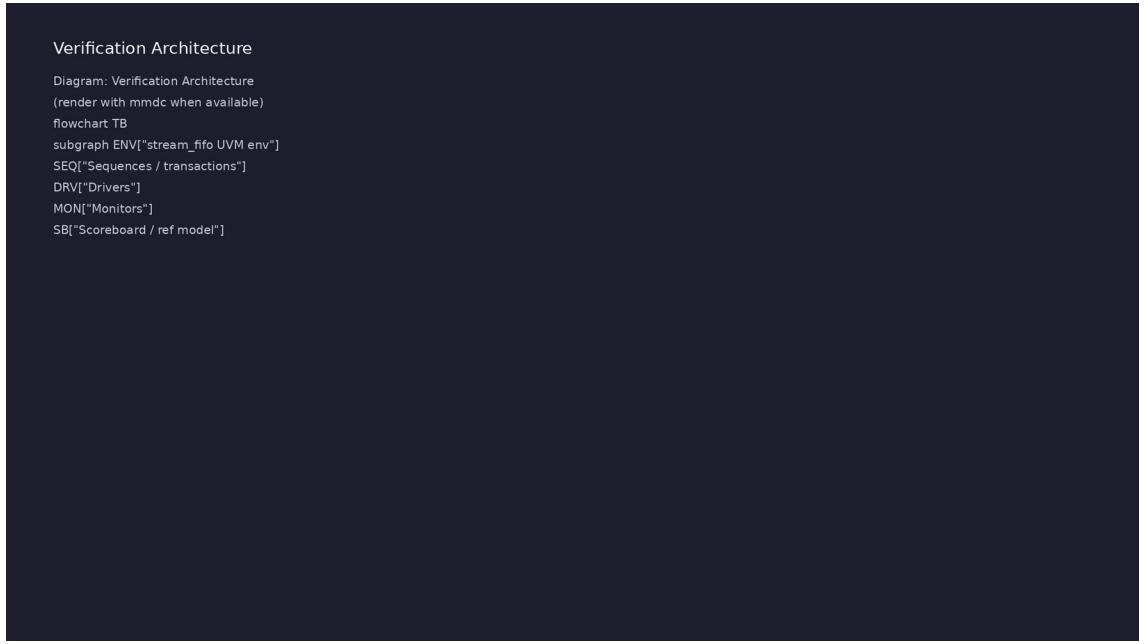
Design architecture

1. Mature UVM environment (stream_fifo)

- Env (`stream_fifo_env`):
  connects source/sink agents, scoreboard, coverage subscribers.
- Agents: driver + sequencer + monitor per interface; active vs passive roles documented in `ENV_DESI…
- Transactions: item fields mirror DUT handshake semantics (data, valid/ready timing).
- Reference implementation sketch:
  `module4/tb/tb_stream_fifo_uvm.sv` and `stream_pkg.sv` patterns.

2. Checking architecture

- Scoreboard / reference model — golden data path and ordering checks (`CHECKER_PLAN.md`).
- SVA — protocol and functional assertions on interfaces and internal flags.
- Clear split: what is checked in SVA vs scoreboard vs test self-check.



DUT instantiation in UVM top

```
// DUT instance
  stream_fifo #(
    .DATA_WIDTH (DATA_WIDTH),
    .DEPTH      (DEPTH)
  ) dut (
    .clk       (clk),
    .rst_n     (rst_n),
    // source side
    .s_valid   (s_if.valid),
    .s_data    (s_if.data),
    .s_ready   (s_if.ready),
    // sink side
    .m_valid   (m_if.valid),
    .m_data    (m_if.data),
    .m_ready   (m_if.ready),
    // status
# ...
```

stream_env — build & connect

```
class stream_env extends uvm_env;

  `uvm_component_utils(stream_env)

  stream_driver     drv;
  stream_monitor    mon;
  stream_scoreboard sb;

  function new(string name, uvm_component parent);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    drv = stream_driver    ::type_id::create("drv", this);
    mon = stream_monitor   ::type_id::create("mon", this);
# ...
```

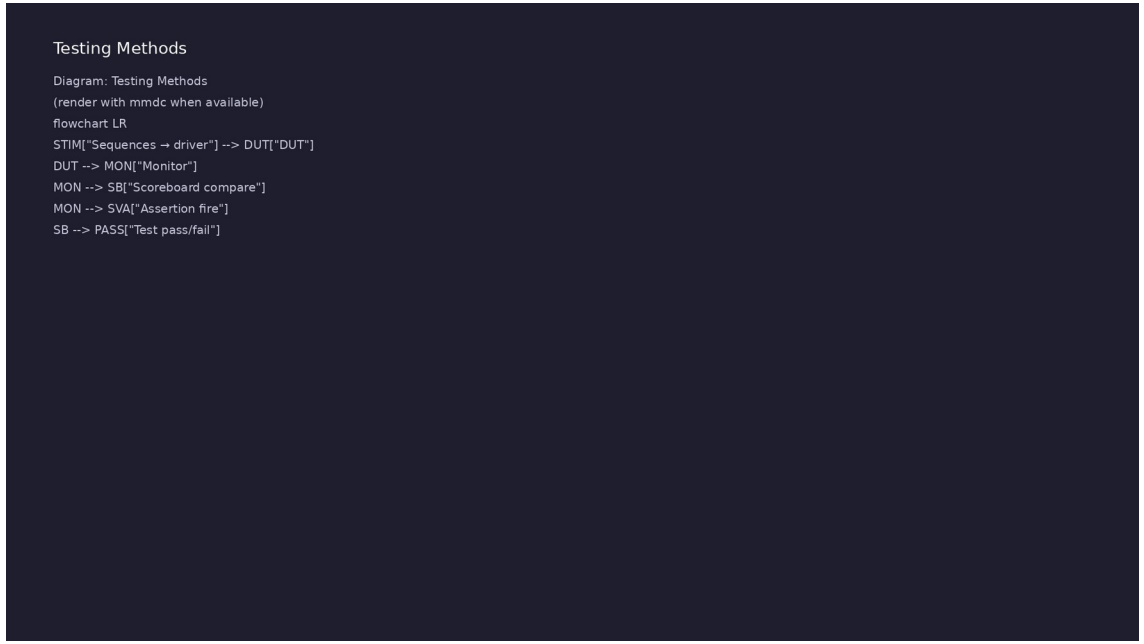
Scoreboard write() hook

```
class stream_scoreboard extends uvm_component;

  `uvm_component_utils(stream_scoreboard)

  uvm_analysis_imp #(stream_item, stream_scoreboard) imp_in;

  int unsigned num_seen;

  function new(string name, uvm_component parent);
    super.new(name, parent);
    imp_in = new("imp_in", this);
    num_seen = 0;
  endfunction

  // Called for each transaction seen by the monitor
  function void write(stream_item t);
# ...
```

Verification & testing methods

1. Stimulus and checking flow

- Sequences → driver → DUT → monitor → scoreboard compare and assertion sampling.
- Reset, backpressure, and error-injection scenarios exercised per `TEST_PLAN.md`.



2. Assertion-based verification (ABV)

- Place assertions close to interfaces; use `assert`/`assume` policy consistent with your simulator f…
- Document waiver and severity policy for known benign cases.

3. Environment review

- Peer/self review of `ENV_DESIGN.md` and `CHECKER_PLAN.md` before expanding regression scope.
- `./scripts/module4.sh --check` validates design docs and TB file presence.

Syllabus topics

1. Environment and Agent Architecture

- Environment and agent architecture (env vs agents vs tests).
- Layered testbench structure and reuse across tiers and DUT variants.

2. Transactions and Sequences

- Transaction and sequence design aligned with your test plan.
- Mapping test intents to UVM sequences and configuration.

3. Drivers and Monitors

- Driver and monitor behavior, including resets, backpressure, and error handling.

4. Scoreboards and Reference Models

- Scoreboard and reference model design.

5. Assertions and Checking Strategy

- Assertion strategy: what to check with SVA vs scoreboards vs tests.

Hands-on examples

Module 4 self-check

```
$ ./scripts/module4.sh --help
```

Terminal: module4_check

```
[0;36m==========================================[0m
[0;36mModule 4: UVM Environment & Checker Maturity[0m
[0;36m==========================================[0m

Usage: ./scripts/module4.sh [OPTIONS]

Module 4: UVM Environment & Checker Maturity (SV/UVM)
This script summarizes and checks the planning artifacts for Module 4.

OPTIONS:
    --env-status          Show status of env_design.md only
    --checker-status      Show status of checker_plan.md only
    --checklist-status    Show status of checklist_module4.md only
    --summary             Show all statuses (default)
    --check               Media/CI self-check (structure only)
    --demo                Print example commands from EXAMPLES.md
    --scaffold            Copy templates/*.md into module4/ if missing
    --help, -h          Show this help message

EXAMPLES:
    # Full summary
    ./scripts/module4.sh

    # Focus only on environment design
    ./scripts/module4.sh --env-status
```

Exercise scaffold

```
./scripts/module4.sh --scaffold
```

Demo: Environment design template

```
$ head -30 module4/templates/ENV_DESIGN.md
```

Terminal: ex01_env_design

```
# UVM Environment & Agent Design (Module 4)

> **Purpose**: Define the detailed UVM environment, agent, transaction, and sequence design for
> **Module**: 4 – UVM Environment & Checker Maturity (SV/UVM)

> **Instructions**: Fill in this document for your design. Copy from `module4/templates/` if nee

## 1. Context

<!-- TODO: DUT name, key interfaces, prior artifacts. Main verification concerns driving env des

## 2. Environment Block Diagram

<!-- TODO: Describe or sketch environment hierarchy (uvm_test_top, env, agents, driver, monitor,

## 3. Component Responsibilities

<!-- TODO: Table of Component, Type, Responsibilities, Notes. -->

## 4. Transaction Models

<!-- TODO: Main transaction classes, key fields, constraints, helper methods, mapping to DUT. --

## 5. Sequence Design

### 5.1 Base Sequences

<!-- TODO: Table of Sequence Class, Purpose/Intent, Uses Transaction(s), Notes. -->

### 5.2 Feature and Stress Sequences
```

Demo: Checker plan template

```
$ head -25 module4/templates/CHECKER_PLAN.md
```

Terminal: ex02_checker_plan

```
# Checker, Scoreboard & Assertion Plan (Module 4)

> **Purpose**: Define the functional checking strategy for the DUT: scoreboards, reference model
> **Module**: 4 – UVM Environment & Checker Maturity (SV/UVM)

> **Instructions**: Fill in this document for your design. Copy from `module4/templates/` if nee

## 1. Context

<!-- TODO: DUT name, key correctness concerns, related documents. Overall checking philosophy (a

## 2. Scoreboards and Reference Models

### 2.1 Scoreboards

<!-- TODO: Table of Scoreboard ID, Component/Class, Inputs, Purpose, Notes. Matching strategy, d

### 2.2 Reference Models

<!-- TODO: Table of Ref Model ID, Implementation, Abstraction Level, Used By, Notes. Inputs, out

## 3. Assertions and Protocol/Functional Checkers

### 3.1 Assertion Inventory
```

Demo: Module-local UVM TB

```
$ head -35 module4/tb/tb_stream_fifo_uvm.sv
```

Terminal: ex03_module_tb

```
`timescale 1ns/1ps

// ------------------------------------------------------------------------------
// UVM-based testbench for the common stream_fifo DUT
// ------------------------------------------------------------------------------
//
// This top-level testbench:
// - Instantiates the shared DUT: common_dut/rtl/stream_fifo.sv
// - Instantiates two stream_if interfaces (source and sink)
// - Connects interfaces to DUT ports
// - Puts the virtual interfaces into the UVM config DB
// - Calls run_test("stream_test") defined in stream_pkg.sv
//

`include "uvm_macros.svh"
import uvm_pkg::*;

// Import the UVM components for this bench
`include "stream_if.sv"
`include "stream_pkg.sv"

module tb_stream_fifo_uvm;

  localparam int DATA_WIDTH = 8;
  localparam int DEPTH      = 4;

  // Clock and reset
  logic clk;
  logic rst_n;

  // Source and sink interfaces
  stream_if #(.DATA_WIDTH(DATA_WIDTH)) s_if (.*);
  stream_if #(.DATA_WIDTH(DATA_WIDTH)) m_if (.*);

  // Status from DUT
```

Demo: Reference env design

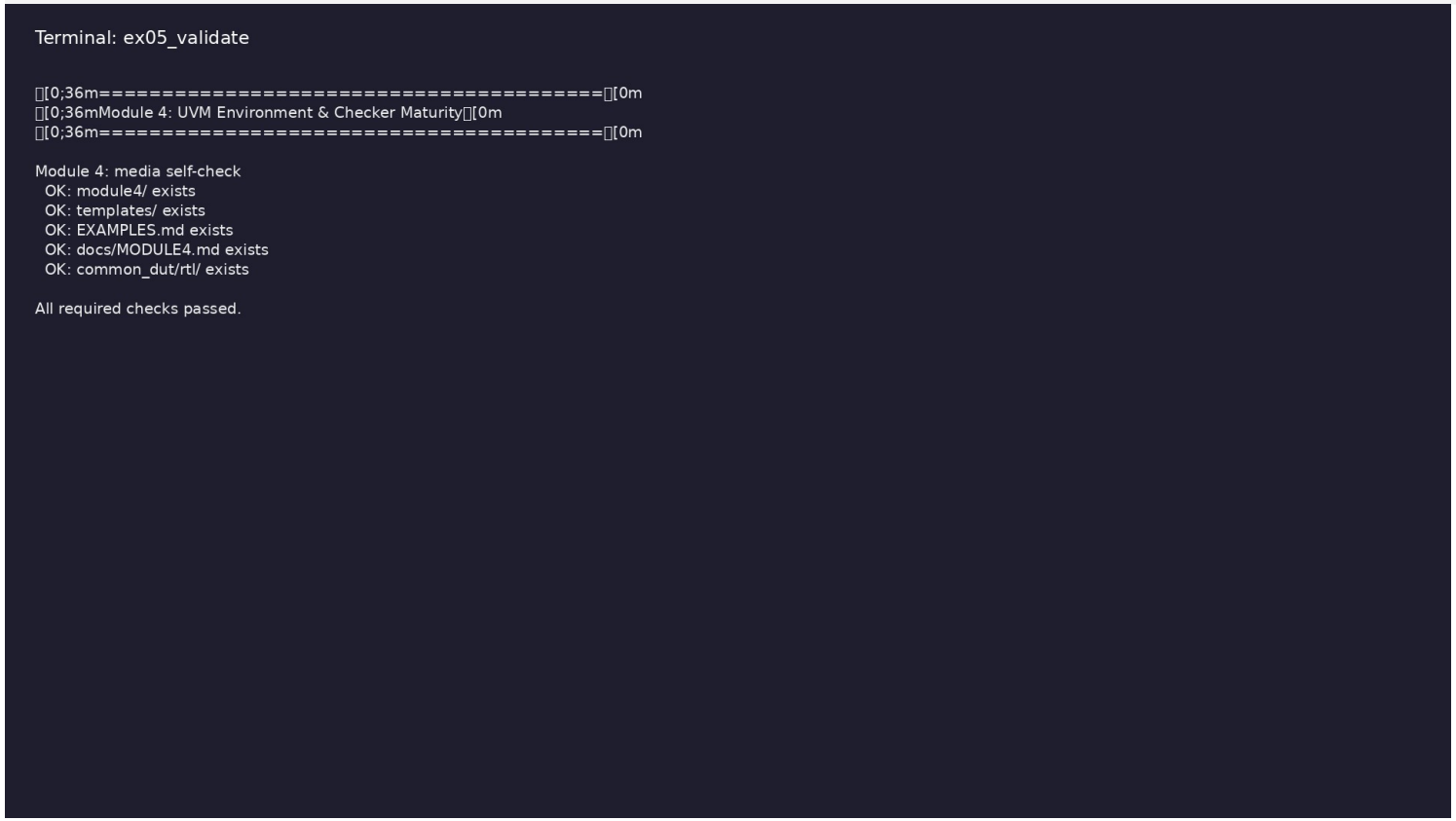
```
$ head -20 module4/.solutions/ENV_DESIGN.md
```

Terminal: ex04_solutions

```
# UVM Environment & Agent Design (Module 4)

> **Purpose**: Define the detailed UVM environment, agent, transaction, and sequence design for
> **Module**: 4 – UVM Environment & Checker Maturity (SV/UVM)

## 1. Context

- **DUT**: `stream_fifo` — parameterizable streaming FIFO with ready/valid handshakes on source
- **Key Interfaces / Protocols**: Source interface (s_valid, s_ready, s_data); sink interface (m
- **Prior artifacts**:
  - `module1/VERIFICATION_PLAN.md` (high-level env & strategy).
  - `module2/TEST_PLAN.md` (test catalogue and types/tiers).
  - `module3/COVERAGE_DESIGN.md` (coverage placement and IDs).

Summarize the **main verification concerns** that drive your env design (e.g., throughput, order

- **In-order transfer (R1)**: Scoreboard must compare popped data with expected order (reference
- **Overflow/underflow flags (R2, R3)**: Assertions and scoreboard must verify flags set only wh
- **Backpressure and protocol**: Driver and monitor must respect valid/ready handshake; tests mu
```

Demo: Module validation

```
$ ./scripts/module4.sh --check
```



Practice & assessment

What you should know (1/2)

- Describe and justify your UVM environment and agent architecture.
- Implement and maintain drivers, monitors, sequencers, and transactions that match your DUT and test…
- Implement or integrate scoreboards and reference models for robust checking.
- Design and place protocol and functional assertions that support your verification goals.
- Configure your environment for different modes and regression tiers.
- UVM env/agent architecture documented in `module4/ENV_DESIGN.md`.

What you should know (2/2)

- Transaction and sequence designs documented and aligned with `module2/TEST_PLAN.md`.
- Scoreboards and reference models specified in `module4/CHECKER_PLAN.md`.
- Key protocol/functional assertions listed and placed.
- At least one env/checker review (self or peer) completed; action items captured.

Assessment checklist

- UVM env/agent architecture documented in `module4/ENV_DESIGN.md`.
- Transaction and sequence designs documented and aligned with `module2/TEST_PLAN.md`.
- Scoreboards and reference models specified in `module4/CHECKER_PLAN.md`.
- Key protocol/functional assertions listed and placed.
- At least one env/checker review (self or peer) completed; action items captured.

Summary & next steps

- Design and mature your **UVM environment** (env, agents, drivers, monitors, sequencers) and **check...
- Complete module4/CHECKLIST.md
- Review module4/EXAMPLES.md and run each lab
- Next: Next module in course